

OBITUARY

Herbert Sichel dies at the age of 79

Herbert Simon Sichel, a *Fellow of the American Statistical Association*, died on September 4, 1995 at his home in Rishon Letzion, Israel. He was 79. With his passing, the statistical sciences have lost an active and innovative researcher and teacher.

Sichel was born in Hanover, Germany in 1915. In 1935 he matriculated *cum laude* at the Leibnitz School and, after a brief spell as a voluntary worker, continued his studies at the Technische Hochschule Berlin Charlottenburg. After only three semesters his studies were rudely interrupted by his arrest for activities as a scout leader in Nazi Germany. He was released in the expectation that he would lead authorities to others, but succeeded in escaping the country and emigrated to South Africa in February 1937. Sichel was subsequently condemned to death in absentia by the Nazi regime.

The University of Pretoria recognized his educational credentials and he was able, language barriers notwithstanding, to complete a B.Sc. degree with distinction in Applied Mathematics and Physics in only two years. In 1950 he obtained his M.Sc. for the dissertation *Mine Valuation and Maximum Likelihood* and in 1953 he was awarded the D.Sc. for the thesis *Contributions to Mathematical Statistics*, both from the University of the Witwatersrand.

In 1939 Herbert Sichel started his career as an underground sampler in the gold mining industry. This early experience gave him a deep insight into the

statistical, physiological and operational problems associated with mining, and laid the foundation for much of his subsequent research, since it introduced him to the exciting world of applied statistics. His first publication in 1947, *An Experimental and Theoretical Investigation of Bias and Error in Mine Sampling with Special Reference to Narrow Gold Reefes* had great influence on mining in many countries. In South Africa it led to a change in government policies towards certain mining practices. It was translated into several languages, including Russian and Japanese and was the seminal work of a new discipline, Geostatistics. It also set the pattern for his lifelong interest in theoretical work motivated by real world challenges and the development of practical tools for solving them. His contributions, which are widely quoted, included “Sichel’s t-statistic”, today a standard tool in the evaluation of lognormally distributed ore reserves, and the Generalised Inverse Gaussian-Poisson Distribution (known as the “Sichel Distribution”) which is commonly used in modelling long-tailed discrete data in a wide range of disciplines, including the evaluation of diamondiferous deposits, actuarial science, library circulation data analysis, species distribution and linguistics.

After eight years in the gold mining industry Sichel moved to the National Institute of Personnel Research in South Africa where he founded its Statistics Division. The nature of the practical problems he sought to explain shifted to psychometrics and psychological research, ranging from highly technical statistical applications, to aptitude and selection testing and absenteeism. The work he did here led to the co-authorship of a book, *Accident Proneness*. During

this period he spent a year at the Educational Testing Service at the University of Princeton, where he shared the company of several eminent mathematicians, including the aging Einstein, Deming and Shewhart.

Herbert Sichel has been described as the father of operations research in South Africa and did more than anyone else to establish statistics and operations research in mining, government, industry and commerce in that country. His pioneering work continued with the founding, in 1952, of his consulting company Operational Research Bureau, which he directed for 19 years. During this time he introduced scientific data processing and data-based decision making to numerous large organizations, establishing operations research and statistics sections in many of these companies and training and guiding their internal specialists. His influence was multiplied many times over by those who undertook their first projects under his tutelage. His expertise in statistical problem solving was in much demand in a wide range of fields, including deep level environmental health hazards in mining, queueing problems in open-cast mines, sampling and ore evaluation with respect to the gold and uranium content of underground gold bearing reefs, the density and sizing of diamond deposits, preventative maintenance in industry and methods to provide a constant mix of ores to iron and steel blast furnaces to increase their efficiency. At the same time, his publications in scientific journals continued to flow at a rate of up to five per year.

While in private practice Sichel lectured for two years on a part-time basis at the newly formed Graduate School of Business Administration at the University of the Witwatersrand. In 1971 he accepted the full-time position of professor in Statistics and Operations Research at this institution, and was also acting director for a two-year period (1971–73). His sabbatical was spent as Lady David Visiting Professor at the Technion-Israel Institute of Technology.

After retirement from the Graduate School of Business Administration at the end of 1980, he was appointed Helen and Norman Asher Visiting Professor at the Technion (1981–83) and Adjunct Professor (1983–84). During the latter period he was also Visiting Full Professor at the University of Haifa. In 1984 he returned to the University of the Witwatersrand where he held the position of Visiting Professor in the Department of Statistics and Actuarial Science until 1991. This was followed by an appointment as Honorary Professorial Research Fellow at the same institution, a position he held from 1992 until his final retirement in 1994. During his sojourn in academia, Sichel continued to consult to industry, notably to De Beers, while his field of research activity expanded to include statistical linguistics, bibliometrics, actuarial science, management decision making and market research. As a teacher and academic, insisting on high standards of academic excellence first and foremost, he was held in highest esteem and was asked by his students, year after year, not to retire. Professor Sichel was always ready to cooperate with his colleagues, to motivate his students and to pass on some of his vast store of knowledge, insight and experience. Today, over 50 post-graduates living all over the world, many of

whom achieved doctoral degrees, are the product of his professional advice and his perpetual encouragement.

Sichel's desire to spread his enthusiasm for his discipline was seen early on in his efforts as the motivating force behind the founding of the Operations Research Society of South Africa. He was its first president and was elected an honorary life member in 1986. He was a fellow, past president and founder of the South African Statistical Association where he was elected honorary life member in 1984 and honorary life president in 1992. He was a fellow of the South African Institute of Mining and Metallurgy which awarded its gold medal to him on three occasions over a thirty year period (1959, 1972, 1988) – a rare distinction. In 1993 he was elected honorary life fellow of the South African Geostatistics Association. In addition, he was a fellow of the American Statistical Association, a fellow of the Royal Statistical Society, and a member of the International Statistical Institute, the Israel Statistical Association and the Psychometric Society (USA).

Herbert Sichel was a well-known statistician, admired for his innovative contributions to science and his ability to communicate to his students and to professional audiences. He was an invited lecturer at universities in several countries and presented papers at national and international congresses and learned societies in South Africa, the United States, Israel, England, Austria, Australia, Canada and Zambia.

Sichel had over 70 published research papers to his credit, most recently (May 1995) he co-authored *New Generalized Model of Observed Ore Value*

Distributions in Transactions of the Institution of Mining and Metallurgy (London), the same journal in which his very first paper had appeared 47 years before. His love for his work drove him to continue his research and consulting after his retirement. On the afternoon of his passing, he made final edits to the paper, *Kurtosis and the Generalised Compound Normal Distribution* which appeared in *The Mathematical Scientist* in 1995.

Academic recognition of Sichel's work was bestowed in 1977 when the University of South Africa conferred on him its honorary degree of Doctor of Science and in 1982 when the University of the Witwatersrand conferred on him the honorary degree of Doctor of Economic Science.

Notwithstanding his achievements and honours, Herbert Sichel remained a humble and self-effacing person. All who knew him will remember him as a dedicated scientist and a wonderful friend. Addressing his colleagues on one occasion, he reminded them that "we stand on the shoulders of giants." He is acknowledged as having been a giant in his own right, a true pioneer who made a lasting contribution.

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